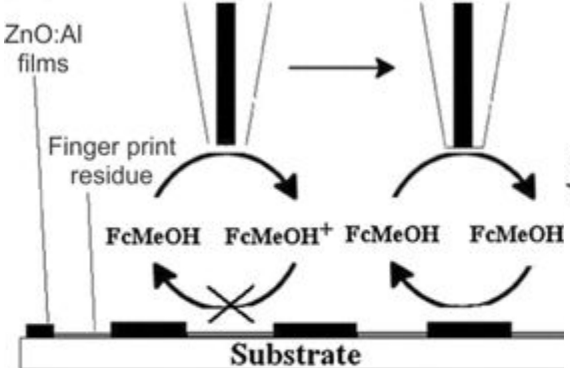
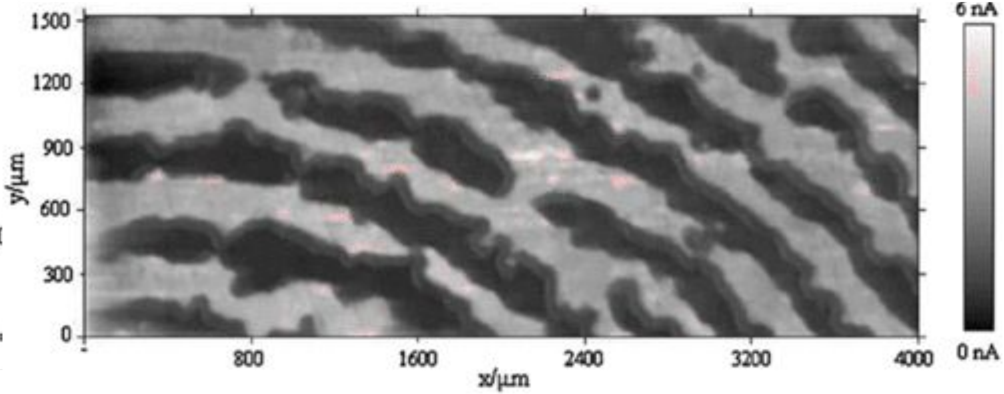
Scanning Electrochemical Microscopy (SECM)

Presenter: Rachel Lee

Ch452

Presented May 6, 2026





What is Scanning Electrochemical Microscopy (SECM)?

- Scanning probe technique introduced in 1989 by Allen J. Bard
- Measures the local electrochemical activity of a sample in solution
- Can be used to investigate the kinetics of reactions (especially in samples which are not easily measured by bulk electrochemistry), locate active sites, and much, much more!

Scanning Electrochemical Microscopy. Introduction and Principles

Allen J. Bard,* Fu-Ren F. Fan, Juhyoun Kwak, and Ovadia Lev

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Bard, A. J.; Fan, F. R. F.; Kwak, Juhyoun.; Lev, Ovadia. Scanning Electrochemical Microscopy. Introduction and Principles. *Anal. Chem.* 1989, *61* (2), 132–138. https://doi.org/10.1021/ac00177a011.

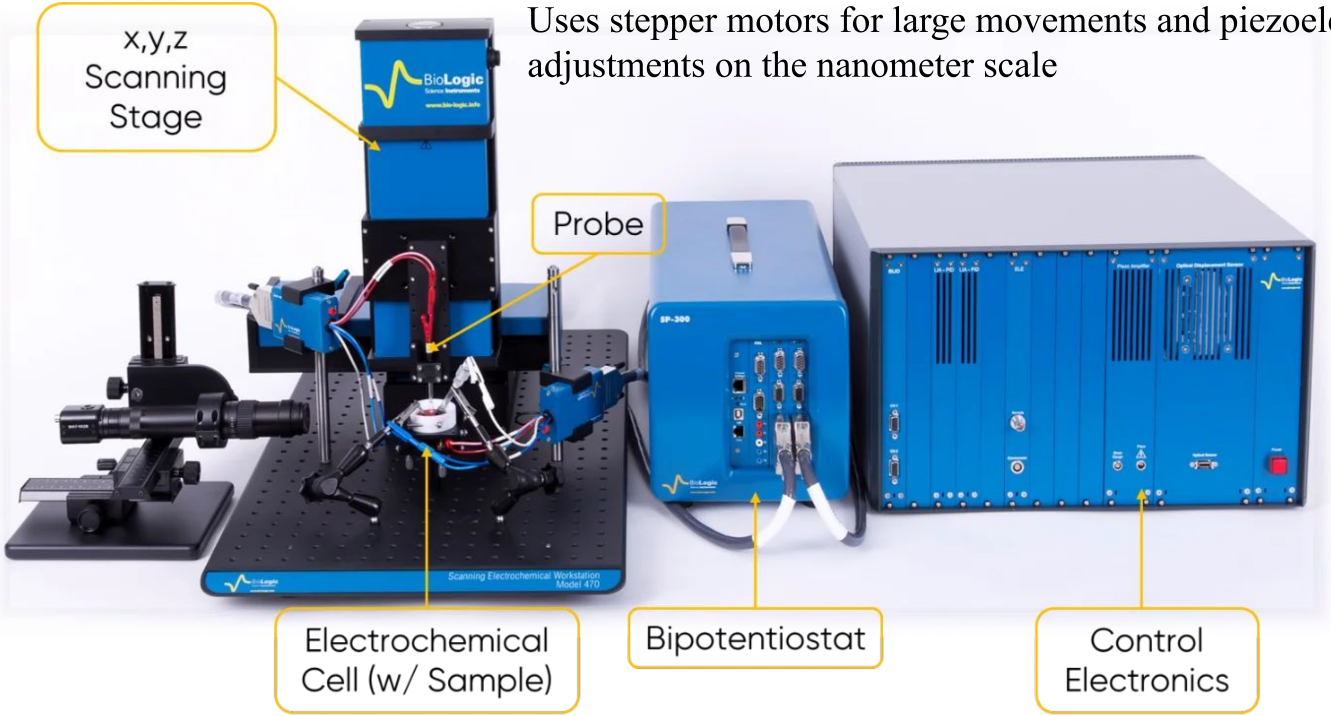
How does SECM work?

Bipotentiostat

Control and measure the potential and current at the probe and substrate

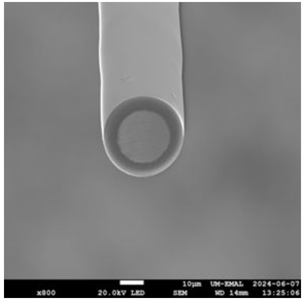
Spatial Positioning System

Uses stepper motors for large movements and piezoelectric motors for positional adjustments on the nanometer scale



SECM101: An Introduction to Scanning Electrochemical Microscopy - BioLogic Learning Center. BioLogic. https://www.biologic.net/topics/secm101-an-introduction-to-scanning-electrochemical-microscopy/ (accessed 2026-05-04).

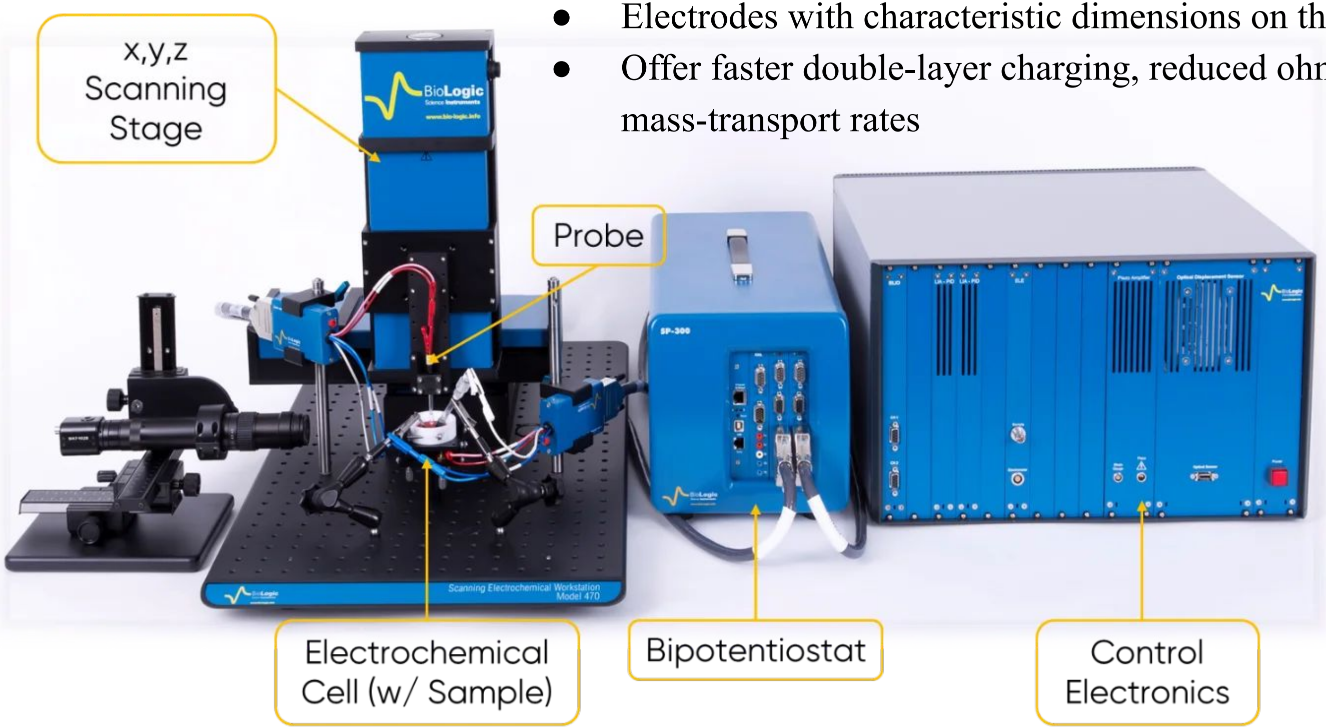
How does SECM work?

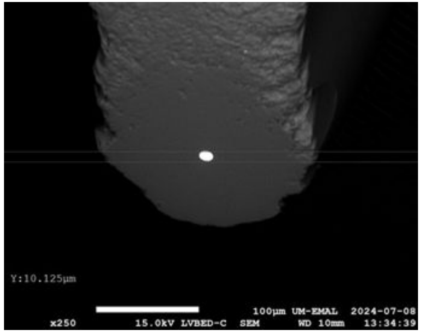


Probe

Usually use ultramicroelectrodes (UMEs) as probes

- Best practices: UME has diameter of 25 µm or smaller
- Electrodes with characteristic dimensions on the (sub) micrometer scale
- Offer faster double-layer charging, reduced ohmic loss, and high mass-transport rates





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Theoretical Background

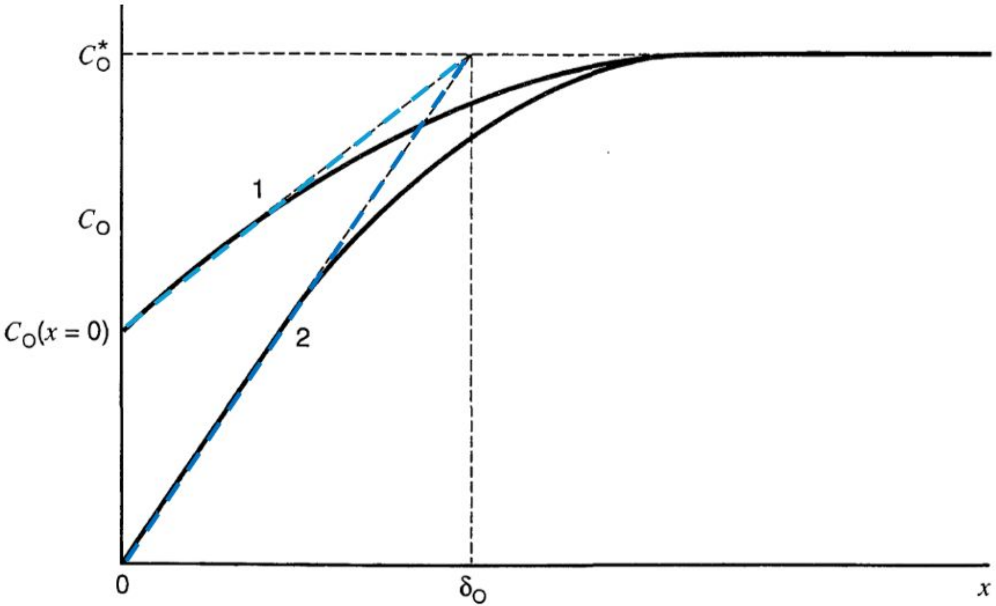


Figure 1.4.1 Concentration profiles (solid lines) and diffusion layer approximation (dashed lines). $x = 0$ corresponds to the electrode surface and δ_O is the diffusion layer thickness. Concentration profiles are shown at two different electrode potentials: (*1*) where $C_O(x = 0)$ is about $C_O^*/2$, (*2*) where $C_O(x = 0) \approx 0$ and $i = i_l$.

Recall:

- Mass transport!
  - Transport mechanisms include diffusion, migration, and convection
- Diffusion controlled reactions have limiting currents expressed as

at the limiting current... $C_o(x=0) = 0$

$$\bar{i} = \boxed{i_l = nFAm_o C_o^*} \quad (1)$$

"maximum rate"

where $m_o = D_o/\delta$, C_o^* is the bulk concentration, and A is the area of the electrode.

How does SECM work?

- UME is immersed in the solution (containing electrolyte and electroactive species)
- Let the electroactive species be initially in its oxidation state, Ox
- When a sufficiently negative potential is applied to the UME, the reduction of Ox occurs at the UME at a diffusion-controlled rate, creating cathodic current through the UME
- The current decays as a diffusion layer of Ox builds up around the electrode and attains a steady state value, $i_{T,\infty}$, described by

$$i_{T,\infty} = 4nFDaC^*$$

where a is a radius of the UME and C* is the concentration of Ox

Doesn't this look familiar?

at the limiting current... $C_0(x=0) = 0$

$$i = \boxed{i_\ell = nFAm_0 C_0^*} \quad (1)$$

"maximum rate"

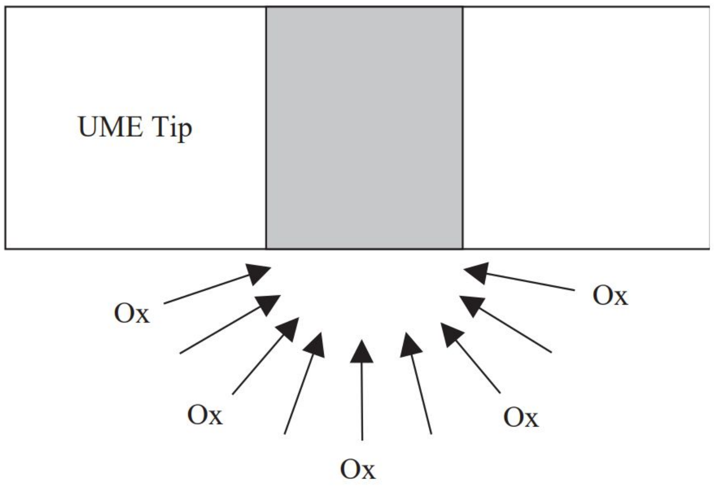
$$i_\ell = nFA\frac{D_0}{\delta}C_0^*$$

$$i_{T,\infty} = 4nFDaC^*$$

where a is a radius of the UME and C* is the concentration of Ox

Difference because we're looking at hemispherical diffusion, not 1D anymore!

Hemispherical Diffusion



Doesn't this look familiar?

at the limiting current... $C_o(x=0)=0$

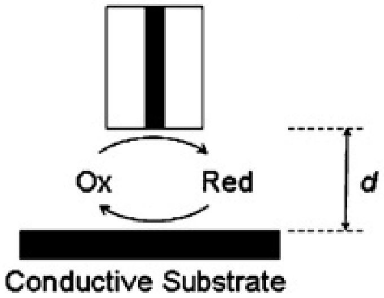
$$\bar{i} = \boxed{i_l = nFAm_o C_o^*} \quad (1)$$

"maximum rate"

$$i_l = nFA\frac{D_O}{\delta}C_o^*$$

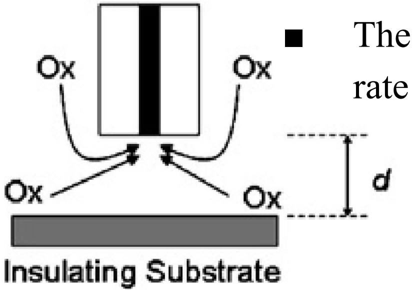
Handbook of Electrochemistry; Zoski, C. G., Ed.; Elsevier Science: Amsterdam Boston, Mass. Heidelberg, 2007.

Data Collection



Feedback mode is the simplest and most common form of SECM

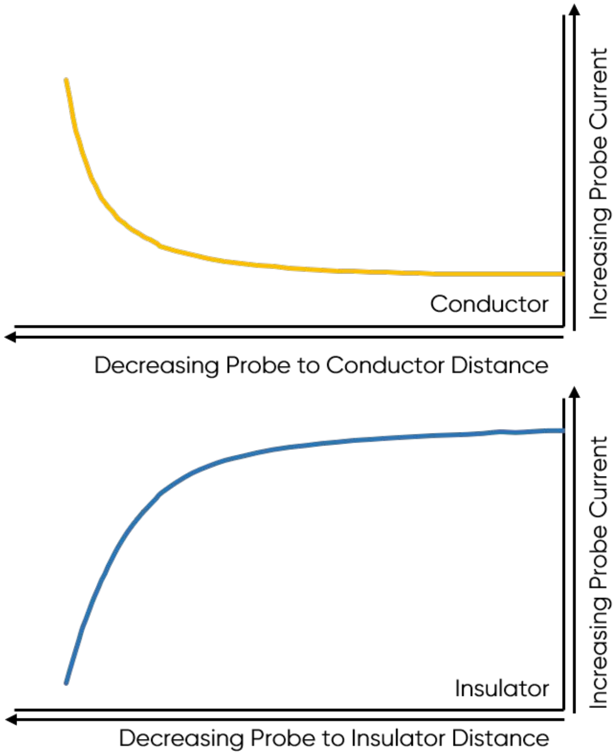
- A redox mediator is introduced into the solution
- The bipotentiostat biases the probe so that the probe reacts with the redox mediator
- Probe-substrate distance large: only a steady state Faradaic current is measured
- Probe-substrate distance small:
    - Insulative/non-reactive substrate: **negative feedback**
        - When the probe tip gets closer to the substrate surface the solid substrate blocks diffusion of the redox mediator to the electrode tip
    - Conductive/active catalyst substrate: **positive feedback**
        - The substrate regenerates the redox mediator. As the probe tip approaches the substrate surface, the rate of reaction increases



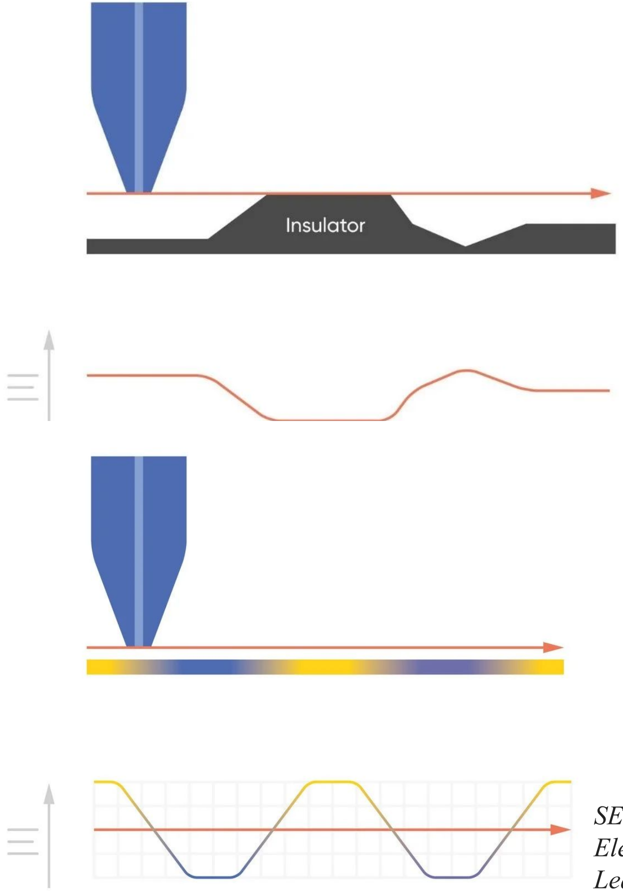
Note: other modes (eg. generation-collection mode and redox competition mode) exist!

Handbook of Electrochemistry; Zoski, C. G., Ed.; Elsevier Science: Amsterdam Boston, Mass. Heidelberg, 2007.

Data Collection



Probe current with decreasing probe-substrate distance over a conductor vs. over an insulator.



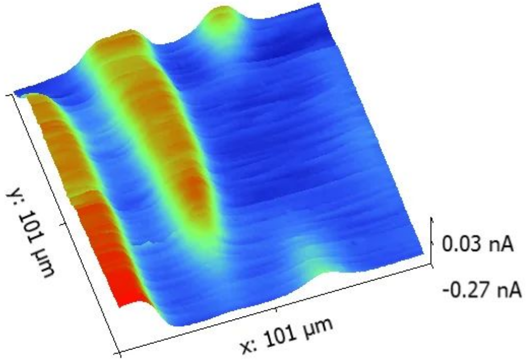
Current as probe moves over an insulative substrate with varying topography.

Current as probe moves over a substrate with varying chemical activity.

SECM101: An Introduction to Scanning Electrochemical Microscopy - BioLogic Learning Center. BioLogic. https://www.biologic.net/topics/secm101-an-introduction-to-scanning-electrochemical-microscopy/ (accessed 2026-05-04).

Applications of SECM

- SECM approach curve
  - Measure probe response as a function of probe-substrate distance
  - Have been used to measure sample swelling over time and under changing sample bias conditions
  - Investigate the kinetics of reactions occurring at samples which are not easily measured by bulk electrochemistry, for example living cells and 2D materials.
- Line scan
  - Used to build of area scan of sample (probe signal plotted as a function of x-y position)
  - Useful for visualizing the local electrochemical features of a sample, performed over time can be used to follow the evolution of electrochemical processes

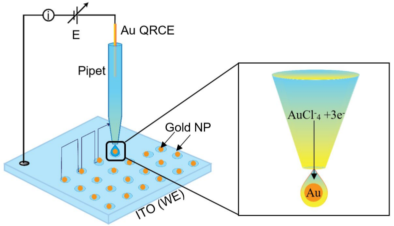


SECM area scan of a spider plant leaf measured in 0.1 x 10^{-3} mol L^{-1}. The 1 µm Pt probe was biased at -0.75 V to measure O_2.

SECM101: An Introduction to Scanning Electrochemical Microscopy - BioLogic Learning Center. BioLogic. https://www.biologic.net/topics/secm101-an-introduction-to-scanning-electrochemical-microscopy/ (accessed 2026-05-04).

Applications of SECM

- Advantages:
  - Able to locally detect sample electrochemical activity (no longer limited bulk measurements)
  - Non-contact
    - Biological applications (ion transport across channels, enzyme activity)
    - Fragile samples (thin films)
    - Liquid-liquid interfaces
  - Highly adaptable and applicable
    - Can be used as imaging devices, electrochemical tool to study surface reactivity of thin films, high-resolution fabrication tool
    - Can be used to measure ANYTHING (insulator? No problem. Conductor? No problem, either!)
- Disadvantages:
  - As the probe is immersed in the bulk solution, there is no way to prevent background noise and bulk interference
  - An aside: this is solved by SECCM (scanning electrochemical cell microscopy)!



Fabricating well-ordered Au nanoparticle arrays for biosensing platforms via scanning electrochemical cell microscopy. https://acs.digitellinc.com/p/s/fabricating-well-ordered-au-nanoparticle-arrays-for-biosensing-platforms-via-scanning-electrochemical-cell-microscopy-543328 (accessed 2026-05-06).

Example Application

Proc Natl Acad Sci U S A. 2013 Jun 4;110(23):9249-54. doi: 10.1073/pnas.1214809110.
Epub 2013 May 17.

Assessment of multidrug resistance on cell coculture patterns using scanning electrochemical microscopy

Sabine Kuss [1], David Polcari, Matthias Geissler, Daniel Brassard, Janine Mauzeroll

Affiliations + expand

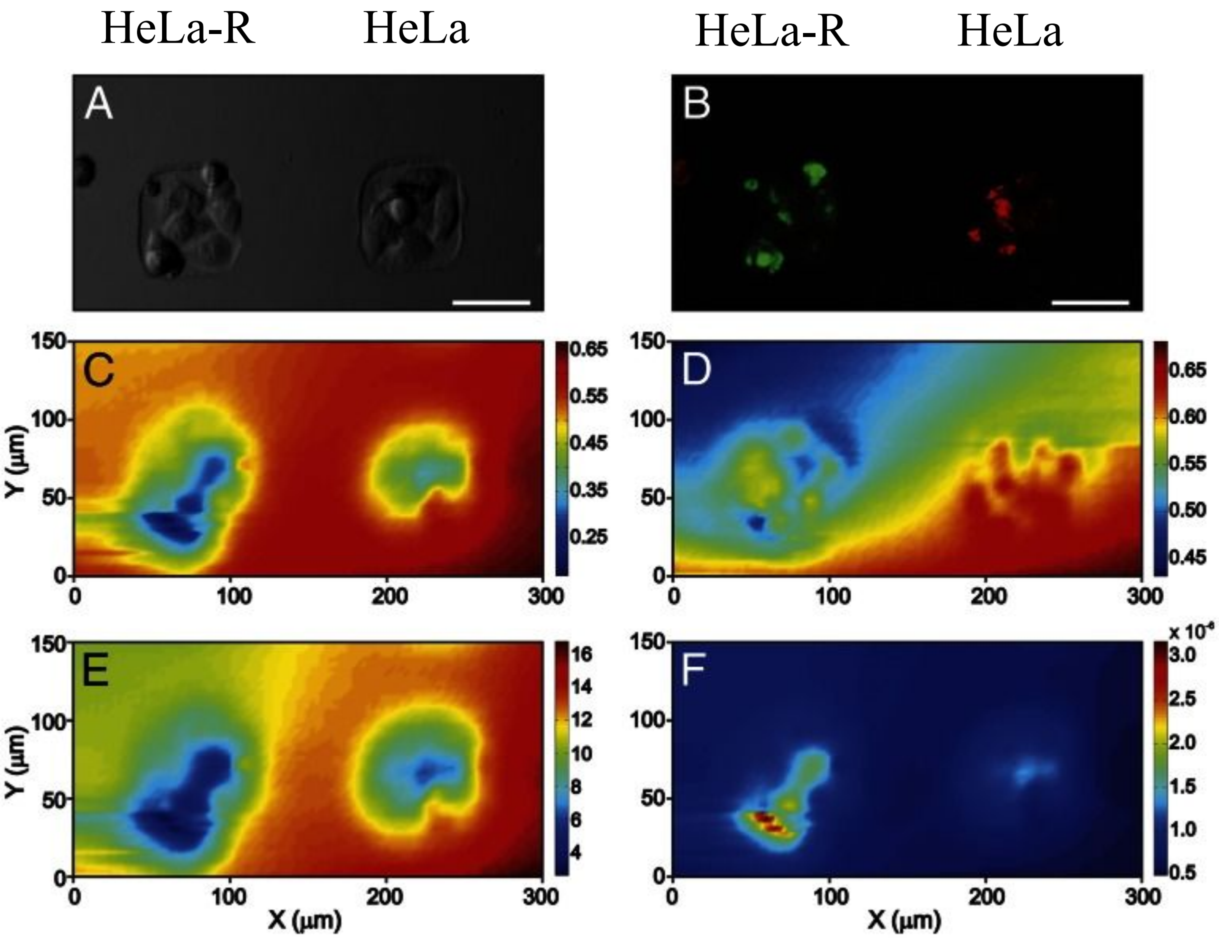
PMID: 23686580 PMCID: PMC3677433 DOI: 10.1073/pnas.1214809110

- Drug resistance is a significant hurdle in developing chemotherapeutic drugs
- Resistance usually driven by overexpression of transmembrane proteins (MRP1) which remove therapeutic agents from the cell
- How can we track transporter activity on a single-cell level without interfering with the cell's natural processes (metabolism)?
- SECM is a wonderful non-invasive method to monitor MRP1 activity in real time!

Example Application

A: optical images
B: fluorescent images
C: SECM currents recorded with the same sample at 12 μm above the substrate in 1 mM $[Ru(NH_3)_6]^{3+}$ (cell-impermeable) (Pt WE, Ag/AgCl RE)
D: same as C but in 1 mM $FcCH_2OH$ (cell-permeable)
E: Normalized tip-to-substrate distance profile
F: Profile of the extracted apparent heterogeneous rate constant ($cm \cdot s^{-1}$) for the sample shown in *A*. (Scale bar: 50 μm.)

Results:
Clear difference in current responses between HeLa-R and HeLa cells (HeLa-R MRP1 activity 2.4 times greater than that in the non-resistant cells)



Kuss, S.; Polcari, D.; Geissler, M.; Brassard, D.; Mauzeroll, J. Assessment of Multidrug Resistance on Cell Coculture Patterns Using Scanning Electrochemical Microscopy. *Proc. Natl. Acad. Sci. U.S.A.* 2013, *110* (23), 9249–9254. https://doi.org/10.1073/pnas.1214809110.

In Summary

- SECM works by immersing a UME in solution and measuring the current as redox-active species diffuse between the substrate surface and the electrode
- This method is used to scan structural features and measure the chemical reactivity of substrates locally
- Highly versatile method with many applications

References

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